Luca Tomarelli

Full Stack & Applied AI Engineer

Berlin · lucatomarelli1@gmail.com · +49 1734287628 · LinkedIn · Portfolio · GitHub

PROFILE

Full Stack Developer with 5 years of experience shipping production web and mobile apps, now specialising in Applied AI: RAG systems, AI agents and machine learning.

I own features end to end, from product and technical decisions to implementation, testing and deployment. Core stack: TypeScript, React, React Native, Next.js and Node.js, plus Python, FastAPI, scikit-learn and LLM APIs. I work closely with product managers, designers and backend teams to turn ideas into production-ready features.

EXPERIENCE

Applied AI Upskilling & Freelance Development
01/2026 – Present | Berlin

- **AI Engineering bootcamp**, SPICED Academy / neue fische (06/2026 – present): full-time, eight months of Python, machine learning and agentic AI. Before it, self-study in RAG and LLM applications (01/2026 – 05/2026) with TypeScript.
- Built and deployed client websites with Next.js.
- **Docs Diver** (live, GitHub): RAG workspace for PDFs. Structure-aware chunking, pgvector semantic search, a token-budgeted chunk selection policy, streamed answers with persisted citations, and per-request observability (latency, tokens, similarity scores). Next.js, Prisma, PostgreSQL, OpenAI, Vercel AI SDK.
- **OpenArt, bootcamp capstone** (live, GitHub): full-stack app that finds open calls for artists, checks eligibility and helps draft applications.
  - Built a multi-label eligibility classifier on ~700 hand-annotated sentences (bge embeddings + logistic regression).
  - Shipped with Next.js, FastAPI, PostgreSQL/pgvector, local ONNX embeddings for RAG and matching, an OpenAI drafting agent with human review, and Docker/nginx deployment.

Front-End Engineer — Sonnen GmbH
01/2023 – 10/2025 | Berlin

Renewable energy company; mobile app used by tens of thousands of active users.

- Developed and shipped the production mobile app in TypeScript and React Native.
- First in-house mobile developer: ran technical interviews and helped grow the front-end team to more than 15 engineers.
- Worked with IoT and backend teams in a cross-functional task force on a new generation of EV chargers.

Full Stack Developer — Northdale UG
06/2021 – 06/2022 | Berlin

- Key developer of Noloni, a video conferencing app for iOS and Android (TypeScript, React Native, Node.js backend on Linux).
- Built supporting React web apps: a management dashboard with metrics and a customer care tool for the support team.

Front-End Developer — MDSLab
01/2021 – 06/2021 | Remote

- Implemented the web and mobile UI of Doeasy, a multi-vendor e-commerce platform, with React and React Native.

SKILLS

AI & ML: RAG, LLM APIs, AI agents, LangChain / LangGraph, embeddings and semantic search, pgvector, scikit-learn, TensorFlow, pandas, NumPy, model evaluation

Frontend: TypeScript, React, React Native, Next.js

Backend & Data: Python, FastAPI, Node.js, Express, REST APIs, GraphQL, PostgreSQL, Prisma, SQLAlchemy

DevOps & Tools: Docker, nginx, GitHub Actions, Fastlane, Linux, Git

EDUCATION

AI Engineering — Data Science & Machine Learning

SPICED Academy / neue fische | *06/2026 – Present* | *Berlin*

Full-time, eight-month bootcamp: Python, machine learning, agentic AI (scikit-learn, TensorFlow, LangGraph, Docker).

Full Stack Web Development

SPICED Academy | *04/2020 – 07/2020* | *Berlin*

Three-month bootcamp: TypeScript, React, Node.js.

B.A. in Political Science and International Relations

Università degli Studi Roma Tre | *09/2013 – 06/2016* | *Rome*

Thesis: Privacy and Data Protection in the European Union.

LANGUAGES

Italian (native) · English (C1) · Spanish (B1) · German (A2)